

YAN-CHENG (BILL) HSU

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EDUCATION

M.S., Computer Science (Artificial Intelligence), UC San Diego 2021 – 2023
B.S., Electrical Engineering, National Central University, Taiwan 2015 – 2019

EXPERIENCE

Site Reliability Engineer II Jul 2025 – Present

Alibaba Cloud U.S. — AnalyticDB Org, AI Training Platform • Sunnyvale, CA

Cross-Cluster AI Training Infrastructure for Unitree G1-D

- Designed **Dual-Layer Virtual Kubelet** architecture orchestrating **10,000+ heterogeneous GPUs**; enabled Serverless-to-Reserved migration achieving **40% cost reduction** and **25x training capacity** for Unitree G1 humanoid robots.
- Engineered **Federated Identity Mesh** enabling cross-cluster AuthN/AuthZ via external Secret injection; resolved split-brain identity challenges with Custom Controllers implementing automated 9-hour token rotation.
- Automated VPC provisioning via **Terraform** (Security Groups, CLB ACLs) to bypass CoreDNS isolation; re-architected **Ray service discovery** for low-latency cross-boundary communication.
- Built **Kubernetes Controller** with telemetry pipeline (**DCGM metrics**, kernel traces); implemented reconciliation loop triggering automated cordon/drain for faulty GPU nodes.

Software Development Engineer Oct 2023 – Jul 2025

Amazon — AGI Org, High Performance Computing • Seattle, WA

Platform Leviathan: NVIDIA A100/H100 Infrastructure for Amazon NOVA

- **GPU Lifecycle & Blocklisting**: Architected scalable tracking system handling **3,000+ concurrent scaling requests**; designed parent-child DAG in Apache Airflow; eliminated circular termination loops, delivering **\$9M annualized savings**.
- **Automated Reliability Orchestrator**: Designed fault identification workflow for **7,000+ GPU** fleet using divide-and-conquer with dynamic K8s node labeling; reduced troubleshooting by **90%** (10h → 1h), saving **1,200 engineer-hours/year**.
- **API & Monitoring**: Built customer-facing API/CLI (Lambda, Smithy); optimized DynamoDB $O(N) \rightarrow O(\log N)$, improving response by **70%**; deployed CloudWatch monitoring achieving **100% failure detection**.

SKILLS

Languages: Golang, Python, C/C++, Java, TypeScript, SQL

AI Infrastructure: NVIDIA A100/H100, Ray (Cluster/Serve), PyTorch Distributed, DCGM, NCCL

Cloud & Kubernetes: Kubernetes Internals (Operators, CRDs, Virtual Kubelet), AWS EKS, Helm, Docker

IaC & Observability: Terraform, AWS CDK, CloudFormation, Prometheus, Grafana, eBPF, Apache Airflow

PUBLICATIONS

Hsu, Y.C. et al. "Generalized DNN Model for Cuffless Blood Pressure Estimation." Sensors 2020

Latifa, N.; Hsu, Y.C. et al. "Optimal Self-Supervised Multi-Fault Detector for Temperature Sensor Data." APSIPA 2023

Amazon AGI et al. "The Amazon Nova Family of Models: Technical Report." 2024